

$$\begin{array}{ccc}
 U & \xrightarrow{f} & U & U \subset S \\
 \varphi \downarrow & & \uparrow \varphi^{-1} & \\
 \varphi(U) & \xrightarrow{z \mapsto ze^{i\theta}} & \varphi(U) & \varphi(U) = \mathbb{D} \subset \mathbb{C}
 \end{array}$$